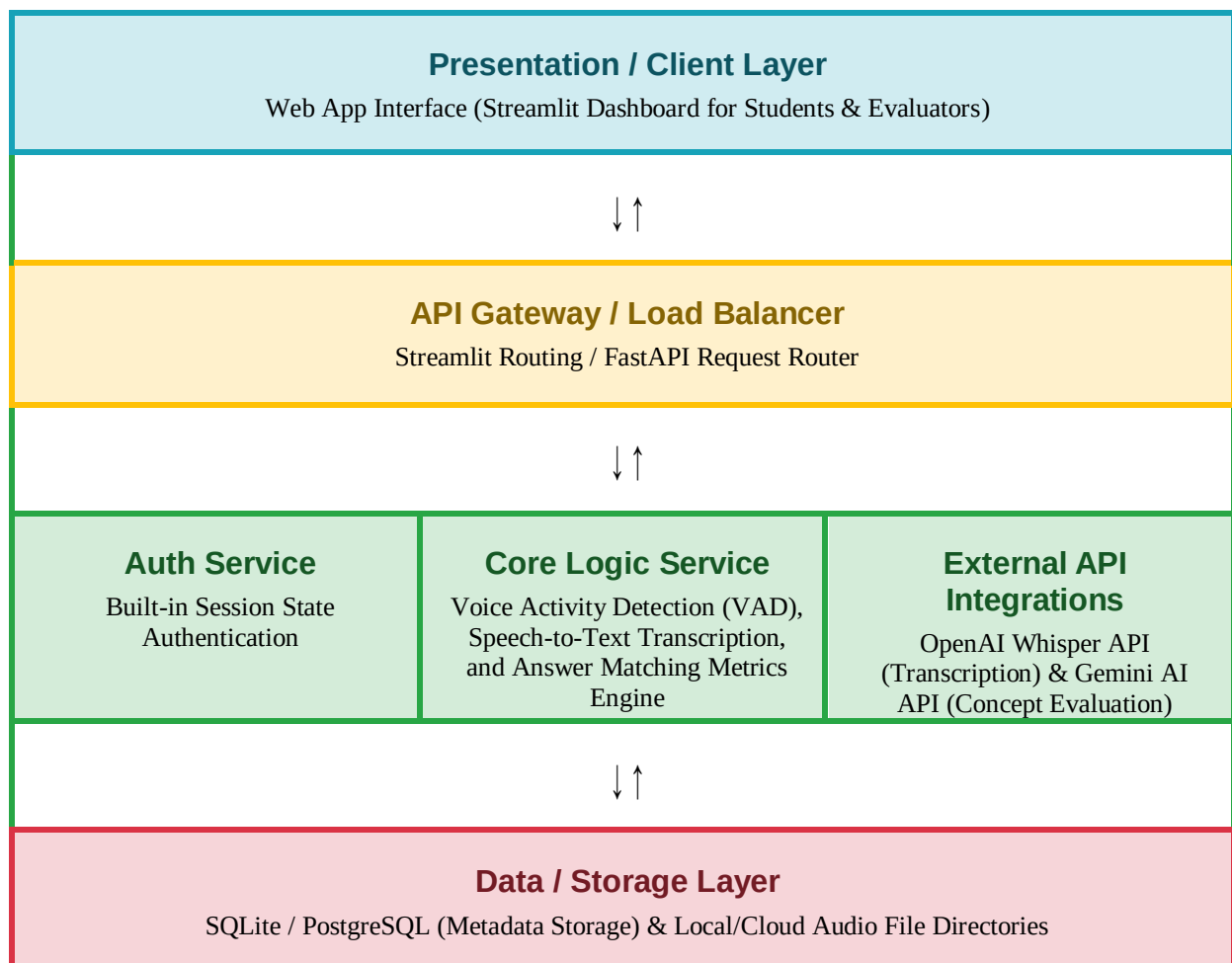


Solution Architecture

Date	28 June 2026
Team ID	XXXXXX
Project Name	Voice-Based Concept Understanding Analyser
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for students to record audio and view reports, and for professors to view the analytics dashboard.	Streamlit (Python)
API Gateway	Handles frontend requests, manages routing to backend logic, and connects to external processing microservices.	FastAPI / Streamlit Routing
Microservices	Manages core tasks: processing raw audio files, transcribing text via Whisper, and grading technical accuracy with an LLM.	Python, OpenAI Whisper API, Gemini AI API
Database	Securely stores student credentials, calculated fluency scores, text transcripts, and feedback logs.	SQLite / PostgreSQL